

Lab 3, Act 1

Trusting a positive detection

NRES 746 – name: _____ group: _____

The scenario

A regional wildlife agency is monitoring a rare forest carnivore using motion-triggered camera traps. Most trigger events aren't the target species at all (deer, rabbits, blowing branches, and hikers set off the cameras more often than the target animal. To keep field biologists from having to review every single image by eye, the agency uses an automated image classifier that flags each trigger event as either “target species detected” or not.

The classifier isn't perfect. For this exercise, suppose:

Quantity	Value
Prevalence: fraction of all trigger events that are actually the target species	1 in 10,000 (0.0001)
Sensitivity: $P(\text{flagged} \mid \text{actually the species})$	0.98
False positive rate: $P(\text{flagged} \mid \text{not the species})$	0.01

That sounds like a pretty good classifier, since it catches 98% of true detections and only mistakenly flags a ‘background’ image not containing the species 1% of the time. **If the classifier flags an image, how worried should the field team be that it's a false alarm?**

Part 1: Compute it with Bayes' rule (individual, ~15 min)

Let S = “trigger event is actually the target species” and F = “classifier flags the image.” You want $P(S \mid F)$.

1a. Compute the two joint probabilities you'll need: $P(F \cap S)$ and $P(F \cap \neg S)$.

1b. Use the law of total probability to combine your two joint probabilities from 1a into the unconditional probability of a flag, $P(F)$.

1c. Now apply Bayes' rule to compute $P(S | F)$ – that is, the probability that a flagged image is a true detection.

1d. Is this number close to what you expected given the sensitivity and false-positive rate reported above? In a sentence or two, what would you tell the field team about how to handle a single flagged image?

Part 2: The same answer, a different way (individual, ~5-8 min)

Bayes' rule can feel abstract. Natural frequencies – working with whole counts of events instead of probabilities – often make the same calculation feel more concrete. Suppose the cameras log 100,000 trigger events over a season. Fill in the table below using the prevalence, sensitivity, and false-positive rate given above (round to the nearest whole number where needed).

	Flagged	Not flagged	Total
Actually the species			10
Not the species			99,990
Total			100,000

2a. From your completed table, read off $P(S | F)$ directly (true detections among flagged images, divided by all flagged images). Does it match your answer to 1c?

Group discussion and revision (~5 min)

Compare your answers and your natural-frequency table with your group. One thing worth discussing: prevalence is doing almost all the work in this result, more than sensitivity or false-positive rate – does that match your intuition, and can you point to exactly where in the calculation that shows up?

Feel free to revise any of your answers above based on the discussion (it's fine to cross out and rewrite – I like to see your thinking evolve).

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion (“muddiest points”) or topics you want to review or dust off going forward.
